Physics 468/668: Statistical Physics **Spring 2019**

Lecture 13: Phase diagrams

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13.1 Phase diagrams

One of the great successes of thermodynamics and statistical mechanics has been in the study of phase transitions. Condensed materials exist in a variety of phases that depend on thermodynamic parameters such as P, V, T, H, E. Thermodynamics and statistical mechanics can be used to determine the properties of individual phases, and the locations and the characteristics of the phase transitions.

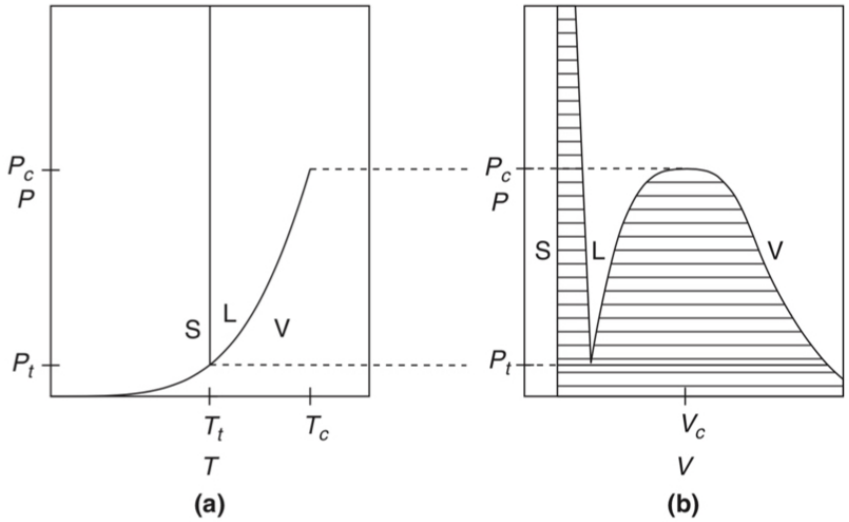


Figure 13.1: $P-T$ and $P-V$ phase Argon.

1. solid, liquid, vapor and supercritical phases
2. critical point
3. triple point
4. coexisting line
5. first-order phase transition
6. discontinuity
7. continuous phase transition

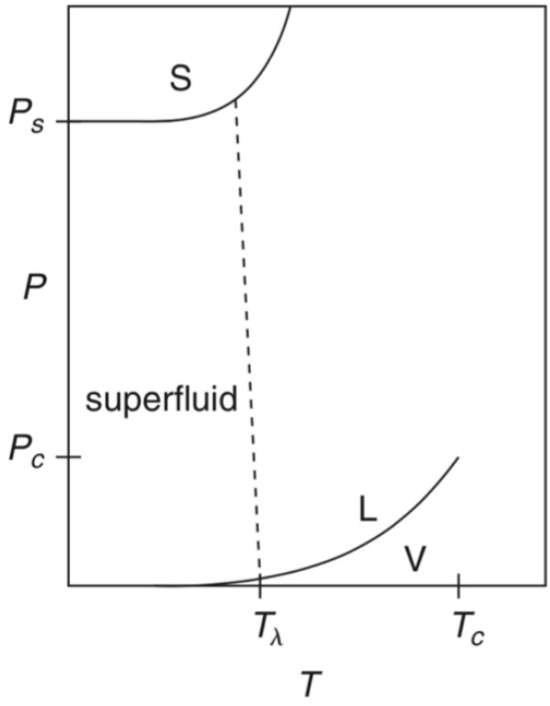


Figure 13.2: $P-T$ phase diagram of He.

1. quantum regime
2. superfluid/supersolid at low $P-T$

13.2 Clausius-Clapeyron equation

The Gibbs free energy of the phases of a material determines the phase diagram.

$$G(N,P,T) = U - TS + PV = A + PV = \mu N \tag{13.1}$$

As the number of molecules in each phase changes, the Gibbs free energy changes by an amount

$$dG = (\frac{\partial G_A}{\partial N_A})_{T,P} dN_A + (\frac{\partial G_B}{\partial N_B})_{T,P} dN_B = (\mu_A - \mu_B) dN_A \tag{13.2}$$

where dN_A is the change of number of particles.

At the equilibrium, G reaches its minimum, so $dG \leq 0$. If $\mu_A > \mu_B$, the number of molecules in phase B will increase and the number in phase A will decrease as the system approaches equilibrium. If $\mu_A < \mu_B$, the number of molecules in phase A will increase and B will decrease. When $\mu_A = \mu_B$, they reach equilibrium.

Let's consider the example of water,

$$\mu_A(P_\delta(T), T) = \mu_B(P_\delta(T), T)) \tag{13.3}$$

The derivatives of the chemical potentials are related by

$$\left(\frac{\partial \mu_A}{\partial T}\right)_P + \left(\frac{\partial \mu_A}{\partial P}\right)_T \frac{dP_\delta}{dT} = \left(\frac{\partial \mu_B}{\partial T}\right)_P + \left(\frac{\partial \mu_B}{\partial P}\right)_T \frac{dP_\delta}{dT} \tag{13.4}$$

The entropy per particle $s = S/N$ and specific volume $v = V/N$ are given by

$$s = -\left(\frac{\partial \mu}{\partial T}\right)_P v = -\left(\frac{\partial \mu}{\partial P}\right)_T \tag{13.5}$$

Therefore,

$$\frac{dP}{dT} = \frac{s_B - s_A}{v_B - v_A} = \frac{\Delta s}{\Delta v} = \frac{L}{T\Delta v} \tag{13.6}$$

Where $L = T\Delta s$ is the latent heat per particle. The slope of the coexistence curve depends on the discontinuities of the entropy per particle and the volume per particle.

At a triple point, the chemical potentials of three phases are equal,

$$\mu_A = \mu_B = \mu_C \tag{13.7}$$

The slopes of the three coexistence lines that define the triple point are related since $\Delta s_{AB} + \Delta s_{BC} + \Delta s_{CA} = 0$ and $\Delta v_{AB} + \Delta v_{BC} + \Delta v_{CA} = 0$. This ensures that each coexistence line between two phases at the triple point points into the third phase. Problem. 4.17

13.3 Quiz

Sketch the $P - T$ and $P - V$ phase diagrams of water, taking into account that the mass density of liquid water is greater than that of solid ice.